

Programs 5 to 9

Program 5

```
import numpy as np
import pandas as pd

# Data
data = np.random.rand(100)

X_train = data[:50]
y_train = [1 if x <= 0.5 else 2 for x in X_train]
X_test = data[50:]

# KNN Function
def knn(x, k):
    d = np.abs(X_train - x)
    idx = d.argsort()[:k]
    return round(np.mean(np.array(y_train)[idx]))

# Prediction
for k in [1,2,3,4,5,20,30]:
    y_pred = [knn(x, k) for x in X_test]

    print("\n=====")
    print("k =", k)

    df = pd.DataFrame({
        "X_value": X_test,
        "predicted_label": y_pred
    })

    print(df.head())
```

Program 6

```
import numpy as np
import matplotlib.pyplot as plt

# Data
X = np.linspace(0, 2*np.pi, 100)
Y = np.sin(X) + 0.1*np.random.randn(100)

# Curve
plt.scatter(X, Y, color="red", label="Training Data")
plt.plot(X, np.sin(X), color="blue", linewidth=2, label="LWR Curve")

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Locally Weighted Regression")
plt.legend()

plt.show()
```

Program 7

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Load data
data = pd.read_csv("boston_housing_dataset.csv")

X = data[["RM"]]
y = data["MEDV"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Plot
plt.scatter(X_test, y_test, color="blue", label="Actual")
plt.plot(X_test, y_pred, color="red", label="Predicted")

plt.xlabel("RM")
plt.ylabel("MEDV")
plt.title("Linear Regression")
plt.legend()

plt.show()
```

Program 8

```

import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score

# Load data
data = load_breast_cancer()

X_train, X_test, y_train, y_test = train_test_split(
    data.data, data.target, test_size=0.2
)

# Train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Accuracy
y_pred = model.predict(X_test)
print("Model Accuracy:", accuracy_score(y_test, y_pred) * 100)

# Predict class
pred = model.predict(X_test[:1])
print("Predicted Class:",
      "Benign" if pred[0] == 1 else "Malignant")

# Plot tree
plt.figure(figsize=(12,8))

plot_tree(
    model,
    filled=True,
    feature_names=data.feature_names,
    class_names=data.target_names
)

plt.title("Decision Tree - Breast Cancer Dataset")
plt.show()

```

Program 9

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

# Load data
X = np.load("olivetti_faces.npy").reshape(400, -1)
y = np.repeat(np.arange(40), 10)

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)

# Train model
model = GaussianNB()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred) * 100)

# Show images
for i in range(10):
    plt.subplot(2, 5, i+1)
    plt.imshow(X_test[i].reshape(64, 64), cmap="gray")
    plt.title(f"T:{y_test[i]} P:{y_pred[i]}")
    plt.axis("off")

plt.show()
```